4039. Sum of Decoded Numbers

Solved

Medium Hint

You are given an integer array `nums`.

Each `nums[i]` is an **encoded** integer representing two positive integers x_i and y_i. To decode `nums[i]`, define:

- $width_i$ = nums[i] % 10 .
- d_i = floor(nums[i] / 10) .
- x_i as the integer formed by the first $width_i$ digits of the decimal representation of d_i.
- y_i as the integer formed by all remaining digits of the decimal representation of d_i.

It is guaranteed that the decimal representation of d_i contains more than $width_i$ digits. Therefore, both x_i and y_i contain at least one digit.

The **decoded value** of `nums[i]` is $x_i^{y_i}$.

Return the sum of the decoded values of all elements in `nums`, modulo $10^9 + 7$.

The `floor()` function returns the integer part of the division.

Example 1:

> **Input:** nums = [231]
>
> **Output:** 8
>
> **Explanation:**
>
> - For 231, we have `width = 1`, `d = 23`, `x = 2`, and `y = 3`.
> - The decoded value of 231 is $2^3 = 8$.
> - Since there is only one element in `nums`, the sum of the decoded values is 8.

Example 2:

> **Input:** nums = [2522,2101]
>
> **Output:** 1649
>
> **Explanation:**
>
> - For 2522, we have `width = 2`, `d = 252`, `x = 25`, and `y = 2`.
> - The decoded value of 2522 is $25^2 = 625$.
> - For 2101, we have `width = 1`, `d = 210`, `x = 2`, and `y = 10`.
> - The decoded value of 2101 is $2^{10} = 1024$.
> - The sum of the decoded values is `625 + 1024 = 1649`.

Example 3:

> **Input:** nums = [2301]
>
> **Output:** 73741817
>
> **Explanation:**
>
> - For 2301, we have `width = 1`, `d = 230`, `x = 2`, and `y = 30`.
> - The decoded value is $2^{30} = 1073741824$.
> - Therefore, the answer is 1073741824 modulo $(10^9 + 7) = 73741817$.

Constraints:

- $1 <= nums.length <= 10^5$
- $100 < nums[i] < 10^{15}$
- $1 <= width_i <= 9$
- $1 <= x_i, y_i < 10^9$
- The digit sequences used to form x_i and y_i do not have leading zeros.
- It is guaranteed that every element in `nums` is a valid encoded integer.

Seen this question in a real interview before? 1/6

Yes No

Accepted 29,649 /66.3K | Acceptance Rate 44.7 %

Topics

Hint 1

Hint 2

Discussion (18)